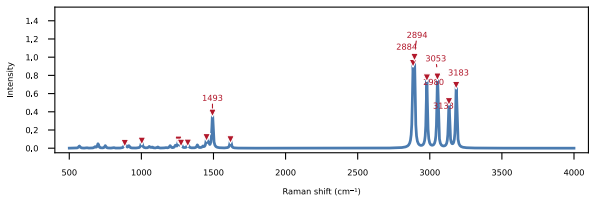
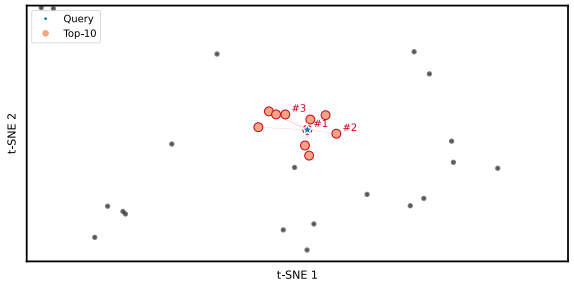
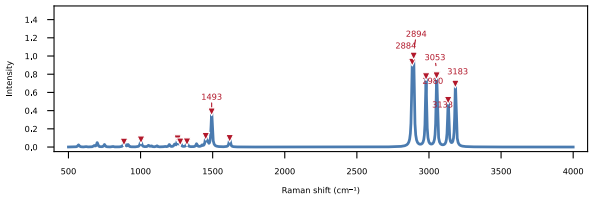
Contrastive Retrieval



Contrastive Encoding



Spectral Transition



Expert Rules & Knowledge Base

- Ih C60: Raman/IR mutually exclusive
- Heavier atoms, lower frequency
- Higher bond order, higher frequency

Guided Reasoning

Candidate Molecules

Rule Matching

Peak Evidence

CoT Reasoning + Checklist Verification

Result :

